Worksheet # 20

Week 10, Day 2: Cluster evaluation, PCA, model evaluation

Name: _____
NetID: _____
Date: _____

Guided checkpoints

For full credit, fill in the checkpoints. If you skip a checkpoint, include your own notes for full credit.

1. Cohesion and Separation

Sketch two clusters for each box.

High cohesion, high separation

Low cohesion, low separation

2. Why Reduce Dimensions?

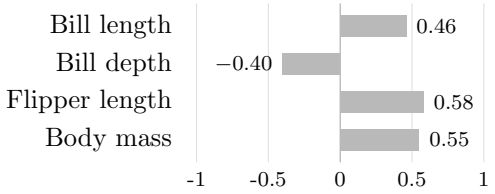
Write two reasons to reduce the number of dimensions in a dataset.

1.

2.

3. Name the Components

PC1 is a weighted sum of all four standardized penguin measurements.



Give PC1 a name: _____

Who came up with that name, you or PCA?

Open-ended notes

4. Precision and Recall

In each matrix, write the name of each cell in the denominator (e.g., TP), then circle the cells in the numerator. Accuracy is done for you.

Accuracy

	pred. +	pred. −
true +	(TP)	FN
true −	FP	(TN)

Precision

	pred. +	pred. −
true +		
true −		

Recall

	pred. +	pred. −
true +		
true −		

Specificity

	pred. +	pred. −
true +		
true −		

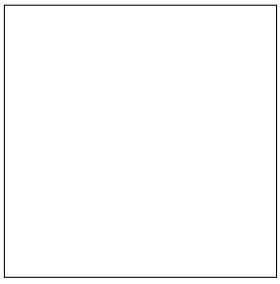
Which cell is ignored by both precision and recall?

Recall is also called: _____

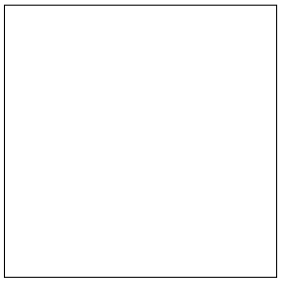
5. Two Curves

Label the axes of each square. Then draw and label three curves in each: a perfect classifier, random guessing, and a "typical" classifier.

ROC



Precision–recall



AUC of the perfect classifier: _____

AUC of random guessing: _____